MEMORY ENHANCING EFFECT OF GOTU KOLA ON MICE EXPOSED TO VANADIUM

BY

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DECLARATION

I hereby declare that this work titled **MEMORY ENHANCING EFFECT OF GOTU KOLA ON MICE EXPOSED TO VANADIUM BY ENELUWE OZICHI CFB/20/21/267444** is an original study that was conducted in the Department Of Pharmacology, Faculty Of Basic Medical Sciences, College Of Health Sciences, Delta State University, Abraka.

ENELUWE OZICHI | **DATE**

(Student signature)

CERTIFICATION

This is to certify that this research work titled **"MEMORY ENHANCING EFFECT OF GOTU KOLA ON MICE EXPOSED TO VANADIUM"** was carried out by **ENELUWE OZICHI,** with matriculation number, **CFB/20/21/267444** in the Department of Pharmacology, Faculty of Basic Medical Sciences, College of Health Sciences, Delta State University, Abraka, in partial fulfillment of the requirements for the award of Bachelor of Science (B.Sc.) Degree in Pharmacology.

DR. ANTHONY T. EDUVIERE **DATE**

(Supervisor)

DR. ANTHONY T. EDUVIERE **DATE**

(Head of Department)

DEDICATION

This work is dedicated to God almighty, the giver of life, wisdom and knowledge and to my amiable parents.

ACKNOWLEDGEMENT

My sincere gratitude goes to God almighty who by in his infinite mercy made this work and journey a success.

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ABSTRACT

BACKGROUND: This study explores the impacts of environmental toxicants on brain health, with a predominant focus on heavy metals and emerging concerns regarding air pollution. It emphasizes that while heavy metal like Vanadium are well-studied, it represent only a fraction of the broader risks posed by environmental toxins found in everyday settings such as food, water, and household products. Chronic exposure to low levels of these toxins is linked to various diseases including cancer, cardiovascular issues, kidney disease and neurodegenerative disease.

Of particular interest is the connection between toxin exposure and neurodegenerative diseases like Alzheimer's and dementia. The study highlights vanadium as a notable neurotoxicant, detailing its adverse effects on the central nervous system through the generation of reactive oxygen species. The research suggests that dietary antioxidants, such as Gotu Kola, may mitigate these effects by enhancing the body's natural antioxidant defenses. Gotu Kola is noted for its potential to protect against neurotoxicity by inhibiting enzymes, preventing amyloid plaque formation, and reducing oxidative stress, thereby offering a promising avenue for therapeutic intervention in neurodegenerative conditions exacerbated by environmental toxins.

METHOD

Study design: The study involved male Albino Swiss mice. Animals were divided into four treatment groups (n = 6 each). Behavioral tests were conducted after 14 days of treatment.

Animal model: Male Albino Swiss mice weighing 24.0±2.0 g were used. Mice were acclimatized for at least one week prior to the experiment. They were housed in standard conditions with controlled temperature and light-dark cycle.

Treatment Protocol: Gotu kola and Vanadium treatments were administered orally and intraperitoneally, respectively. Gotu kola doses were 50mg/kg and 100mg/kg. Vanadium dose was 10mg/kg. Drug administration occurred daily between 9:00am to 10:00 am for 14 days.

Biochemical Analysis: After behavioral tests, mice were euthanized, and brains were collected. Brain tissues were homogenized and centrifuged and Acetylcholinesterase (AChE) activity in brain homogenates was measured spectrophotometrically using DTNB. AChE activity was expressed as μmol/min/g tissue and Glutathione Peroxidase (GPx) activity was also determined spectrophotometrically using hydrogen peroxide (H_2O_2). GPx activity was expressed as U/mL.

RESULT: Gotu Kola demonstrated beneficial effects against vanadium-induced memory impairment in mice across different tests. It improved spatial and recognition memory, reduced acetylcholinesterase activity in the brain (indicative of neuroprotection), and enhanced glutathione peroxidase activity (suggesting antioxidant effects). These findings suggest that Gotu Kola may be a promising therapeutic agent for mitigating neurotoxic effects associated with vanadium exposure.

CONCLUSION: The study concludes that Gotu Kola demonstrates efficacy in reducing memory impairment and neuronal damage induced by heavy metals like Vanadium. This effect is attributed to its ability to inhibit acetylcholinesterase activity, elevate antioxidant levels, and increase glutathione peroxidase. Overall, the findings suggest that Gotu Kola could be therapeutically beneficial in managing and treating memory disorders.

KEYWORDS: Gotu Kola (Centella Asiatica), Vanadium Toxicity, Memory Enhancement, Albino Swiss Mice, Behavioral Tests, Acetylcholinesterase Activity, Spatial Working Memory, Recognition Memory, Neuroprotective Effects, Oxidative Stress

CHAPTER 1

1.0 INTRODUCTION

1.1 BACKGROUND

Studies and documentation have been carried out on the effects of toxicants in the environment on brain and neurological systems. However, most of the research carried out so far has focused on harmful effects from heavy metals and it appears that exposure to these substances represents only a small part of overall environmental health risks. For example, the damaging effects of air pollution on the brain, and particularly on intelligence, are being increasingly viewed as a significant concern. In today's world, we are all exposed to some level of toxins. It is present in our food, water, air and everyday things that we find at home like lotions, washing powder, furniture or building materials (Ugoeze *et al.,* 2021). We're not talking about high dose toxin exposure here; these exposure levels aren't typically enough to lead to acute symptoms. However, studies have shown that low doses may add up over long periods of time and contribute to the development of disease.

Researchers have linked low-level toxin exposure to many diseases including cancer, cardiovascular, and kidney disease (Mitra *et al.,* 2022). Our focus on this research, however, will be on the growing body of medical literature that demonstrates that certain toxins increase the risk of Alzheimer's disease and dementia and how herbal supplement Gotu kola can be use to ameliorate it.

Dementia is a general term for loss of memory/cognitive functioning, language, problem-solving and other thinking abilities that are severe enough to interfere with daily life. Dementia isn't just one disease (Makwana and Elizabeth 2022). A general term to describe the number of symptoms which may occur in people who are living with a variety of diseases, including Alzheimer's disease which is the most common type of dementia. It is a progressive disease beginning with mild memory loss and possibly leading to loss of the ability to carry on a conversation and respond to the environment.

According to researchers, the top environmental toxin categories associated with cognitive decline are heavy metals which include lead, mercury, cadmium, arsenicas as well as vanadium amongst others.

Vanadium (V) is a transition metal that presents in multiple oxidation states and numerous inorganic compounds and is also an ultra-trace element considered to be essential for most living organisms (Özsoy *et al.,* 2022). Despite being one of the lightest metals, Vanadium offers high structural strength and good corrosion resistance and thus has been widely adopted for high-strength steel manufacturing. High doses of Vanadium exposure are toxic, and inhalation exposure to Vanadium adversely affects the respiratory system. The neurotoxicological properties of Vanadium are just beginning to be identified. Recent studies carried out by some researchers demonstrates the neurotoxic potential of this metal in the nigrostriatal system and other parts of the central nervous system (CNS). The neurotoxic effects of Vanadium have been mainly attributed to its ability to induce the generation of reactive oxygen species (ROS) (Aureliano *et al.*, 2023). It is noteworthy that Vanadium is a toxic metal and a potent environmental and occupational pollutant. Since many dietary (exogenous) antioxidants are known to upregulate the intrinsic antioxidant system and ameliorate oxidative stress-related disorders, this review evaluates the effectiveness of Gotu Kola in the treatment of vanadium-induced neurotoxicity. This Gotu Kola has been reported to have a comprehensive neuroprotection by different modes of action such as enzyme inhibition, prevention of amyloid plaque formation in Alzheimer's disease, dopamine neurotoxicity in Parkinson's disease, and decreasing oxidative stress. (Belwal *et al.,* 2019)

1.2 Statement of the Problem

The problem that this study addresses is Cognitive deficit and memory impairement due to heavy metals. The study evaluates Occupational and Environmental toxicity of Vanadium in the brain as well as neuronal system.

Furthermore, it looks into the herbal supplement Gotu kola as a supplementation that can assuage memory impairement induced by Vanadium exposure in mice. Thus, it gives insight on the potential benefits of the supplement as an antioxidant and a neuroprotective agent for ameliorating Vanadium induced memory impairement.

1.3 Aim

The aim of this study is to evaluate the effect of Gotu Kola on memory impairement induced by Vanadium

1.4 General Objectives

The general objectives of the research on the memory enhancing effect of Gotu kola on mice exposed to vanadium are:

- To assess the cognitive performance of mice following exposure to Vanadium through behavioural tests.
- To evaluate the effect of Gotu kola on cognitive/memory functions in mice exposed to Vanadium through behavioural tests.
- To contribute to the understanding of natural remedies for mitigating vanadium-induced memory impairement
- To explore the memory enhancing effect of Gotu kola in the context of Vanadium toxicity

1.5 Specific Objectives

The main objectives of this study include the following:

- To evaluate the effect of Gotu kola on memory in mice exposed to vanadium using the Y-maze.
- To quantify the effect of Gotu kola on memory in mice exposed to vanadium using the Novel Object Recognition Test (NORT).
- To investigate the effect of Gotu kola in oxidative stress biomarkers of mice exposed to Vanadium.
- To evaluate the effect of Gotu kola on Acetylcholinesterase enzyme

1.6 Justification of study

This study addresses the critical need to understand and mitigate the cognitive impairments caused by environmental toxins, specifically vanadium. Despite its industrial utility, vanadium exposure poses serious health risks, including memory loss and cognitive decline. Gotu kola, a traditional herbal supplement, has shown promise in protecting against neurotoxicity through various mechanisms.

1.7 Significance of the study

Vanadium exposure has become a potential environmental and occupational pollutant to human, animal and plant. Its particles are either directly or indirectly ingested or inhaled through food water and air. Upon absorption, studies have shown that it accumulates in the tissues leading to nervous system alterations, paralysis of legs, respiratory failure, convulsions, bloody diarrhea, and death. It also disrupts the blood–brain barrier and alters some neurotransmitters concentrations such as acetylcholine thereby leading to memory deficit, hence the essence of this study which is to evaluate the beneficial role of Gotu kola on Vanadium-induced memory impairment.

1.8 Scope of study

This study focuses on evaluating the memory-enhancing effects of Gotu kola on mice exposed to vanadium. The scope includes:

i. Animal Model: Using mice as the experimental subjects to model vanadium-induced cognitive impairment.
ii. Exposure and Treatment: Administering vanadium to induce neurotoxicity and subsequently treating with Gotu kola to assess its potential protective effects.
iii. Behavioral Tests: Conducting cognitive and memory assessments using the Y-maze and Novel Object Recognition Test (NORT) to measure the impact of Gotu kola on memory functions.
iv. Biomarker Analysis: Investigating oxidative stress biomarkers and acetylcholinesterase enzyme activity in the brains of treated mice.
v. Comparative Analysis: Comparing cognitive performance and neurobiological markers between control, vanadium-exposed, and Gotu kola-treated groups to determine the efficacy of Gotu kola.

CHAPTER 2

2.0 LITERATURE REVIEW

2.1 MEMORY

Memory is the faculty of the mind by which data or information is encoded, stored, and retrieved when needed (Matta *et al.*,.2020). For the purpose of influencing future actions, it is the retention of information over time, (Coulter *et al.*, 2021). In memory, there are three main processes: encoding, storage and retrieval. (Ortega-de San Luis *et al.*, 2022).

The history of psychological studies on memories dated back hundreds of years. Since at least the time of the ancient Greek philosopher Plato, who believed that memory was a blank canvas on which an accurate image of the world was created and maintained indefinitely, people have sought to understand the essence of memory. In 1885, a German psychologist, Hermann Ebbinghanus, carried out the first controlled experiments on memory, and in doing so, he established a pattern for modern experimental studies on memory. Ebbinghanus developed many methods of studying memory that are still in use till date.

2.1.1 TYPES OF MEMORY

In humans, there are basically 3 major types of memory. They include;

- Sensory memory
- Short-term memory
- Long-term memory
- Sensory memory

Sensory memory holds information, derived from the senses, less than one second after an item is perceived. An example of sensory memory is the ability to examine an object and remember what it looks like in a mere second's observation, or memorization (Budson *et al.*, 2022). It is out of cognitive control and is an automatic response (Conway *et al.*, 2024).

There are three types of sensory memories, (Herweg *et al.*, 2020). Iconic memory is a fast-decaying store of visual information, a type of sensory memory that briefly stores an image that has been perceived for a small duration (Wang *et al.*, 2023). Echoic memory is a fast-decaying store of auditory information, also a sensory memory that briefly stores sounds that have been perceived for short durations (Bianco *et al.,* 2023). Haptic memory is a type of sensing memory that's associated with the touch stimulus database (Wan *et al.,* 2020).

- **Short-term memory**

Short term memory refers to a brain system that remembers bits of information for short periods, often less than 30 seconds (Antonijevic *et al.,* 2020). Short-term memory creates a kind of "visuospatial" sketch of information the brain has recently absorbed and will process into memories later on (Forsberg *et al.,* 2021). Some estimates suggest that there are approximately seven items of information held in temporary memory at a time. Sometimes people think of short-term memory and working memory in the same way, but they are not. The memory system that stores pieces of information for a temporary period is referred to as "short-term memory". Working memory is the brain's processes allowing it to manipulate and use information retained (Angelopoulou *et al.*, 2021).

- **Long-term memory**

Storage capacity and duration of sensory memory and short-term memory are generally very limited, (Morey *et al.*, 2020) That is to say that information cannot be stored indefinitely. On the other hand, while the total capacity of long-term memory is still unknown, it can store much more information. Moreover, this information could be kept for a much longer period of time, which may last all the life cycle. For example, given a random seven-digit number, one may remember it for only a few seconds before forgetting, suggesting it was stored in short-term memory. Conversely, it is claimed that a person can recall telephone numbers for long periods of time by repeating them; this information is said to be stored in long-term memory. Long-term memory is grouped into two categories known as explicit memory (declarative memory) and implicit memory (non-declarative memory) (Martin *et al.*, 2021).

Explicit memory refers to all memories that are consciously available (Schneider *et al.,* 2021). These are encoded by the hippocampus, entorhinal cortex, and perirhinal cortex, but consolidated and stored elsewhere. Research by Meulemans and Van der Linden (2003) found that amnesiac patients with damage to the medial temporal lobe performed more poorly on explicit learning tests than did healthy controls. However, these same amnesiac patients performed at the same rate as healthy controls on implicit learning tests. This implies that the medial temporal lobe is heavily involved in explicit learning (McDougle *et al.,* 2022). Explicit memory which is also known as declarative memory is divided into;

Episodic memory which refers to memory for specific events in time (Schneider *et al.,* 2021), as well as supporting their formation and retrieval. Some examples of episodic memory would be remembering someone's name and what happened at your last interaction with each other (Williams*et al.,* 2022).

Semantic memory which refers to knowledge about factual information, such as the meaning of words. Semantic memory is independent information such as information remembered for a test (Beaty *et al.*, 2023).

Autobiographical memory which refers to knowledge about events and personal experiences from an individual's own life. Autobiographical memories are facilitated by aids including verbal, face-evoked, picture-evoked, odour-evoked, and music-evoked autobiographical memory cues. (McCormick *et al.,* 2020). Though similar to episodic memory, it differs in that it contains only those experiences which directly pertain to the individual, from across their lifespan.

Implicit memory (procedural memory) refers to the use of objects or movements of the body, such as how exactly to use a pencil, drive a car, or ride a bicycle (Bellone *et al.,* 2021). This type of memory is encoded, and it is presumed stored by the striatum and other parts of the basal ganglia.

There are four types implicit memory: Procedural, associative, non-associative and priming. Procedural memory is the part of memory that participate in recalling motor and executive skills that are necessary to perform a task (Bellone *et al.*, 2021). It is an essential system that guide ls activity and usually works at an unconscious level. Procedural memories are regained automatically for use, when necessary, in the implementation of difficult procedures related to motor and intellectual skills. Associative memory simply means the storage and retrieval of information through association with other information (Loprinzi *et al.,* 2021). The acquisition of associative memory is carried out with two types of conditioning: The Classical conditioning is associative learning between stimuli and behavior. However, operant conditioning is a form of learning in which new behavior develop based on their consequences. Non-associative memory refers to newly learned behavior through repeated exposure to an isolated stimulus (D’Souza *et* *al.,* 2021). New behavior can be classified into two processes: sensitization and habituation. Priming memory refers to an effect whereby exposure to certain stimuli influences the response given to stimuli presented later (Stewart 2022).

2.2 PARTS OF THE BRAIN INVOLVE IN MEMORY

The main parts of the brain involved with memory are the amygdala, the hippocampus, the cerebellum, and the prefrontal cortex.

- **The Amygdala**

The almond-shaped amygdala is located in the temporal lobe, directly below the uncus. With roughly 13 nuclei divided functionally into five main groups: basolateral nuclei, cortical-like nuclei, central nuclei, additional amygdaloid nuclei, and extended amygdala. The amygdala is a rich and complex structure. The limbic system, which is in charge of controlling emotions and behavior in addition to memory creation, includes the amygdala. The peri-amygdaloid cortex, which is part of the surface of the uncus, merges with the amygdala at the anterior border of the lateral ventricle's inferior horn, which is where the amygdala is located anatomically (Carlo *et al.,* 2021). The anterior choroidal artery supplies blood to the amygdala, and the posterior choroidal vein drains it until it forms the great cerebral vein, which empties into the straight sinus (Govaert *et al.,* 2020). Information processing between the hypothalamus and prefrontal-temporal association cortices is coordinated by the amygdala. The dorsal route and the ventral route are the two main output channels of the amygdala's neuronal circuits, which allow it to perform its various duties. The basal ganglia circuit has been proposed as a substrate for the human ability to deduce the intentions of others from their language, gaze, and gestures (Theory of mind and social cognition). The amygdala also has connections to this circuit through its projections to the ventral striatum and ventral pallidum (Westby 2022). According to Šimić et al. (2021), the amygdala also controls anger, anxiety, fear conditioning, emotional memory, and social cognition. It also plays a role in conditioning when food, sex, and drugs are used as appetite cues. Regarding the amygdala's function in memory, it is known that its activation can modulate the formation and consolidation of memories that cause an emotional reaction (Roesler *et al.,* 2021). Numerous illnesses, including depression (Lu *et al.,* 2022), sleep debt and anger (Schiel *et al.,* 2022), and other neuropsychiatric conditions, have been linked to the amygdala.

- **The Hippocampus**

The hippocampus, sometimes referred to as the "flash drive" of the human brain, is linked to memory consolidation and decision-making; nevertheless, its structure and functions are significantly more intricate than those of a flash drive (Gao *et al.*, 2024). Within the parahippocampal gyrus of the inferior temporal horn of the lateral ventricle, the hippocampus is a convex elevation of gray matter tissue. It resembles a sheet of the cortex that is bent and recurved and folds into the medial surface of the temporal lobe. The dentate gyrus, the hippocampus proper, and the subiculum are the three separate zones that make up the hippocampus. The hippocampal length of five centimeters splits into a head, body, and tail (Yulbarsovna 2024). The fimbria, crus, body, and column are among the components of the hippocampus. The structure that depicts the many levels of the hippocampal formation is called the Cornu Ammonis (CA), and it resembles a seahorse or ram's horn. The four subfields of the hippocampus are called CA1, CA2, CA3, and CA4. On either side of the dentate gyrus's hilus are CA3 and CA2. Dentate granule cells send fibers to CA3, the biggest hippocampal region, via their proximal dendritic branches (Chauhan *et al.*, 2021). The anterior choroidal artery, which extends medially and superiorly to the uncus, supplies blood to the hippocampal region. Information is stored, retrieved, and registered in three stages of memory. Autopsy and imaging investigations have demonstrated the critical role of the hippocampus, parahippocampal region of the medial temporal lobe, and the neocortical association in memory processing. Bilateral injury to the aforementioned regions results in short-term memory impairment that precedes the inability to create new memories (Fernandez-Romero *et al.*, 2021). Because of its strong connections to the amygdala, hypothalamus, septum, and mammillary bodies, the hippocampus is only slightly stimulated by stimulation of these surrounding regions. The hippocampus also sends out a lot of impulses, particularly via the fornix into the anterior thalamus, hypothalamus, and larger limbic system. In addition, the hippocampal region is highly hyperexcitable, which enables it to convert brief electrical stimulation into a prolonged, prolonged stimulation that facilitates the encoding of memories from the touch, visual, auditory, and olfactory modalities (Yulbarsovna 2024). The hippocampus is where judgment calls are made and information is stored for potential use in the future. As a result, it possesses a system that enables short-term memory to be converted into long-term memory, combining verbal and symbolic thought processes into data that can be retrieved for making decisions.

- **The Cerebellum**

The greatest portion of the hindbrain, the cerebellum, is situated in the posterior cranial fossa, behind the medulla oblongata, the pons, and the fourth ventricle (Singh 2020). The cerebellum and cerebrum are separated by the tentorium cerebelli, which is an outgrowth of the dura matter. Eighty percent of the brain's neurons are concentrated in a dense cellular layer in the cerebellum (Nayak *et al.*, 2022). The superior cerebellar artery (SCA), anterior inferior cerebellar artery (AICA), and posterior inferior cerebellar artery (PICA) supply the cerebellum with blood flow. While it cannot start a muscular contraction, the cerebellum is responsible for maintaining posture, coordinating walking, and controlling voluntary muscle activity and continuous muscle contraction (Ertan *et al.,* 2020). Additionally, it synchronizes motor learning and vision. Loss of control over fine motor skills, preservation of balance, and motor learning will occur from damage to this region of the human brain (Marchese *et al.,* 2020).

- **The Prefrontal Cortex (PFC)**

The prefrontal cortex, often known as the "personality center" and the cortical region that gives us our distinct human identity, is one of the last areas of the brain to develop. Because it is not a primarily sensory or solely motor area of the brain, the prefrontal cortex is special. As opposed to this, the prefrontal cortex processes information "in the moment" by receiving input from several cortical regions. Higher-order executive tasks are processed by the PFC through links to other cortical regions. It is feasible to adjust to various situational environments and regulate reactions depending on an individual's perspective at a given time as a result to these reciprocal links. Pro-social skills are better developed in humans than in other primates and non-primate animals because of this ability to correctly integrate incoming information, which is essential for survival during interactions in the social world (Dixon *et al.,* 2022). Numerous neuropsychiatric illnesses, which typically manifest as disinhibition, apathy, loss of initiative, and personality abnormalities, are caused by lesions in this region.

2.3 NEUROTRANSMITTERS INVOLVE IN MEMORY

- **Acetylcholine (ACh)**

Cholinergic neurons release acetylcholine, a neurotransmitter that plays a role in signal transduction linked to memory and learning (Chen *et al.,* 2022). Centrally acting cholinergic drugs are used to treat Alzheimer's disease by taking advantage of the role that cholinergic pathways play in memory (Chen *et al.,* 2022). The encoding of new memories involves both nicotinic and muscarinic receptors. Ach is also demonstrated to increase memory encoding in response to sensory inputs (Udakis *et al.,* 2021). This is caused by the inhibition of excitatory feedback activity in addition to the augmentation of afferent to cortical locations. Presynaptic muscarinic receptors mediate the inhibition of feedback and nicotinic stimulation increase excitatory afferent (Sevilla *et al.,* 2021).

- **Glutamate**

The most prevalent amino acid transmitter in the central nervous system (CNS) involved in learning and memory is the excitatory amino acid glutamate (Du *et al.,* 2020). The mechanism of memory storage known as long-term potentiation, or LTP, is the main focus of glutamate's involvement in memory. N-methyl d-aspartate (NMDA) is the most significant glutamatergic receptor implicated in the production of LTP (Collingridge *et al.,* 2022). Additionally, it has been demonstrated that NMDA receptor activation plays a role in the active process of LTP degradation (Benke *et al.,* 2020).

- **Dopamine**

Dopamine functions as a neurotransmitter in some parts of the brain in addition to being a noradrenaline precursor (Gasmi *et al.,* 2022). Research indicates that dopaminergic neurotransmission plays a crucial role in retrieving learned information from the striatum's long-term memory (Tsetsenis *et al.,* 2023). According to the research, consolidation of spatial memory requires the engagement of both D1 and D2 receptors in the nucleus accumbens (Espadas *et al.,* 2021). Dopamine is also linked to declarative memory that is reward-related and motivated (wang *et al.,* 2023). Nonetheless, it has been demonstrated that D3 receptor antagonism improves memory, attention, and learning by disinhibiting dopamine neurons that project to the prefrontal cortex and boosting Ach release (Kumar *et al.,* 2020).

- **Norepinephrine**

Research has shown that during and after an emotional event, the activation of beta-adrenergic stress hormone systems contributes to the creation of memories linked to emotional arousal (Elsey *et al.,* 2020). Additionally, it has been demonstrated that dopamine and NE both influence prefrontal-dependent working memory and are implicated in the prefrontal area. Additionally, it has been discovered that NE and ACh interact and work together to influence specific memory types (Murala *et al.,* 2022; Cabrera *et al.,* 2024).

- **Gamma-aminobutyric acid (GABA)**

This is the main inhibitory neurotransmitter in the central nervous system (CNS), and it is widely distributed in the areas of the brain related to memory and learning (Kaur *et al.*, 2020). One of the mechanisms underlying the alcohol-induced LTP block, which stops information from moving from short-term to long-term memory, is GABA-A activation (Mira *et al.*, 2020). Data on GABA B also demonstrate its function in a number of assays related to memory, learning, and synaptic plasticity (Sakimoto *et al.*, 2021). Nonetheless, the steady establishment of conditioned inhibition depends on both inotropic (GABA A) and metabotropic (GABA B) receptors (Kaur *et al.*, 2020).

- **Serotonin**

Serotonin is a mood stabilizer and aids in sleep-wake cycles. It is also important in the development of neurodegenerative diseases like Alzheimer's disease because low amounts of the mood regulator may actually drive the disease (Fedrizi *et al.,* 2018).

2.4 MEMORY IMPAIRMENT

Another name for this is amnesia. It describes either total or partial memory loss. Depending on what causes the amnesia, the pathophysiology can differ, however some frequent processes include:

- Damage to Brain Structures: Amnesia may arise from impairment or malfunction in areas of the brain that are essential for storing and retrieving memories. Trauma, stroke, tumors, infections, or neurodegenerative illnesses could all be the cause of this harm. (Mukherjee *et al.,* 2023)
- Hippocampal Dysfunction: The brain's seahorse-shaped hippocampus is essential for creating new memories. When the hippocampal tissue is damaged, either by neurodegeneration or ischemia (blood flow reduction), new memories cannot be created, a condition known as anterograde amnesia.

- Temporal Lobe Dysfunction: The temporal lobes of the brain are involved in memory storage and retrieval. Damage or dysfunction in these areas, including the hippocampus and adjacent structures like the amygdala and entorhinal cortex, can result in various types of amnesia. (Fernandez-Romero and Spica 2021)
- Consolidation Disruption: Memory consolidation is the process by which short-term memories are converted into long-term memories. Disruption of this process, such as through sleep disturbances or certain medications, can lead to difficulties in forming or retaining memories.
- Altered Neurotransmitter Function: Neurotransmitters like acetylcholine, glutamate, and gamma-aminobutyric acid (GABA) play crucial roles in memory processes. Imbalances or dysfunctions in these neurotransmitter systems can contribute to amnesia (Zhang *et al.*, 2023). According to older research from 2014, glutamate is the most common neurotransmitter in the brain. It can cause nerve cells to die through a process known as excitotoxicity. Excitotoxic cell death can cause neurodegenerative conditions that affect memory.
- Psychological Factors: In some cases, psychological factors such as dissociation or repression may play a role in causing amnesia, particularly in response to traumatic events (Mucci 2021).

Understanding the specific pathophysiology of amnesia often involves thorough neurological evaluation, neuroimaging studies, and sometimes neuropsychological testing. Treatment depends on the underlying cause and may include medications, cognitive rehabilitation, psychotherapy, and lifestyle modifications. (Parker *et al.*, 2022)

2.4.1 TYPES OF MEMORY IMPAIRMENT

There are several types of memory impairment but few are reviewed below. They include;

- **Amnesia**

Amnesia refers to the loss of memories, including facts, information and experiences (Langer 2021). There are two types of amnesia, which are;

i. Anterograde amnesia: This refers to a type of memory impairment characterized by difficulties in forming new memories after the onset of a specific event or condition. It primarily affects the ability to encode and retain new information or experiences. In individual with anterograde amnesia, memories formed prior to the onset of amnesia (retrograde amnesia) are generally intact, while the ability to create new memories is impaired. This can include difficulties in remembering recent conversations, events, or learning new skills. The extent and severity of anterograde amnesia can vary depending on the underlying cause and individual factors. (Sparaco *et al.*, 2022).

ii. Retrograde amnesia: This refers to a type of memory impairment characterized by the inability to recall memories or information from the past, usually events or experiences that occurred before a specific event or condition. It is the opposite of anterograde amnesia, where the ability to form new memories is impaired. In retrograde amnesia, the extent and duration of memory loss can vary. The loss of memories is typically more pronounced for events closer to the onset of amnesia, with older memories being relatively more preserved. (Miller *et al.,* 2022).

- **Alzheimer's Disease (AD)**

It is a progressive Neurodegenerative disorder that primarily affects memory, cognition, and behavior. It is the most common cause of dementia, accounting for a significant proportion of dementia case. The disease is characterized by the accumulation of abnormal protein deposits in the brain, specifically beta-amyloid plaques and Tau tangles (Roda *et al.,* 2022). These abnormal protein formation disrupt the normal functioning of brain cells, leading to their degeneration and eventual death. The disease begins with mild memory loss and possibly leading to loss of the ability to carry on a conversation and respond to the environment. Alzheimer's disease involves parts of the brain that control thought, memory, and language.

- **Traumatic Brain Injury (TBI)**

This refers to damage in the brain caused by an external force, such as a blow, jolt, or penetrating injury to the head. TBI can result in a wide range of physical, cognitive and emotional impairments, depending on the severity and location of the injury. TBI can occur as a result of various incidents, including motor vehicle accidents, falls, sport injuries, assaults, and combat-related injuries. The severity of the injury can range from mild (concussion) to severe, with varying degrees of immediate and long-term effects. (Wolfe *et al.,* 2021).

- **Parkinson's Disease**

A neurodegenerative condition that impairs mobility, Parkinson's disease is typified by a range of motor and non-motor symptoms. It is principally caused by the progressive degeneration of dopamine-producing neurons in the substantia nigra, a part of the brain. Parkinson's disease is characterized by four main motor symptoms: rigidity (stiffness of muscle), bradykinesia (slowness of movement), postural inability (poor balance and coordination), and tremor (usually at rest). These symptoms can progressively affect both sides of the body, often starting on one. Non-motor symptoms are another aspect of Parkinson's disease that can manifest either before to or concurrently with motor symptoms.These may include mood disorders (such as anxiety and depression), sleep issues, autonomic dysfunction (such as altered blood pressure and digestion), sensory complaints, and cognitive abnormalities (such as memory and executive function issues). (Weintraub *et al.,* 2022).

2.4.2 MODELS FOR MEMORY IMPAIREMENT INDUCTION

Listed below are few out of many models use in inducing memory impairment:

- Streptozotocin-induced model of AD: Among the alkylating agents of anti-cancer medications is streptozotocin (STZ), a glucosamine nitrosourea molecule. Because it closely resembles human sporadic AD, this is the most commonly used model for AD induction. Increased oxidative and nitrosative stress, tau hyperphosphorylation, brain atrophy, neuronal death, and neuroinflammation are all brought on by STZ infusion (Arora 2020). There was evidence of impaired acetyl coenzyme A, ATP, and creatine phosphate synthesis. Administration of STZ was associated with increased acetylcholinesterase (AChE) activity and apoptosis (Baig and Panchal 2020).
- Amyloid-β model of Alzheimer's disease: Amyloid-β plaques are the primary indicators of this illness. When the Aβ peptide is administered, cognitive impairment is first shown, then neurodegeneration. According to Islamie et al. (2023), intraventricular injection of Aβ into the third ventricle for a period of 14 days causes the buildup of amyloid proteins and plaques in various brain regions, especially in memory-related areas like the hippocampus and cortex, which are followed by the striatum and amygdala. It has been demonstrated

that this model is more appropriate for inducing AD since it causes behavioral changes in addition to changes in biochemicals and neurochemicals (Morgese *et al.*, 2022).

- Scopolamine induced AD model: The well-known anticholinergic medication scopolamine blocks the cholinergic muscarinic receptor, which results in an overabundance of acetylcholine being released and eventually injuring the cortical and hippocampus areas. The medication modifies the cholinergic pathways in the areas of the brain related to cognition and memory (Chen and Yeong 2020; Chen *et al.*, 2022).
- Alcohol-induced memory deficits: It has been established that higher and increased ethanol dosages impair memory by interfering with its encoding, storage, and consolidation processes (Skike *et al.*, 2019). It is well recognized that ethanol affects hippocampal-dependent learning and memory, which ultimately results in cholinergic dysfunction (Macht *et al.*, 2022).
- Phenytoin induced memory deficits: A popular anti-epileptic medication that works well for both partial and tonic-clonic seizures is phenytoin. It is well documented that taking this medication causes blood folic acid levels to drop and causes cognitive deterioration (Novak *et al.*, 2022; Anandan *et al.*, 2024). Administration of phenytoin to rodents caused levels of neurotransmitters like 5-HT to change, and their cholinesterase activity also changed (Rapaka *et al.*, 2022).
- Heavy metals induced memory deficit: Among the commonly used Alzheimerogenic chemicals, heavy metals like aluminum, lead, mercury, Vanadium etc. have been identified to induce high levels of toxicity that is linked with several diseases, including neurodegenerative disorders, following long-term exposure in humans or chronic administration in rodents (Ijomone *et al.*, 2020). Heavy metals neurotoxicity is highly linked with cognitive impairment of AD, via oxidative stress and cholinergic dysfunction (Mahdi *et al.*, 2019).

2.4.3 CAUSES OF MEMORY IMPAIREMENT

Several causes of memory impairment include;

Normal Aging: As people age, their memory may experience mild modifications. Older adults may experience sporadic memory loss or have trouble recalling certain information and thought patterns. According to Dahan et al. (2020), the primary factor contributing to memory impairment in older adults is the age-related degeneration of the hippocampal region, a brain region crucial for memory formation and retrieval. Aging also causes a decrease in the hormones and proteins that support and restore brain tissue as well as promote neural growth (Azam *et al.*, 2021; Radaghdam *et al.*, 2021). Reduced blood flow to the brain is a common occurrence in older adults, which can affect memory and cause changes in cognitive abilities (Bliss *et al.,* 2021).

Neurodegenerative disease: Memory loss may be a sign of neurodegenerative disease such as dementia (Pathak *et al.,* 2021). Dementia also affects thinking, language, judgment, and behavior. Common types of dementia associated with memory loss are:

- Alzheimer disease
- Vascular dementia
- Lewy body dementia
- Fronto-temporal dementia
- Progressive supranuclear palsy
- Normal pressure hydrocephalus
- Creutzfeldt-Jakob disease

Nutritional Deficiency: Memory impairment may result from deficiency in vital minerals or vitamins, such as vitamin B1, B12, A, or D (Kumar *et al.,* 2022).

Alcohol or Drugs of Misuse: Overuse of medications, including hypnotics and barbiturates (Edinoff *et al.,* 2021). According to Lees et al. (2020), alcohol also causes the hippocampal activity to be disrupted.

Sleep Disorder: Memory problems can arise from chronic sleep disorders, such as sleep apnea or insomnia, which interfere with the consolidation of memories (Bottary *et al.,* 2020).

Stroke: According to Nian et al. (2020), a stroke happens when there is an interruption in blood flow to the brain, resulting in brain damage. A stroke can cause memory loss and other forms of cognitive impairment, depending on the part of the brain that is affected (Rost *et al.,* 2022).

Hydrocephalus: This neurological condition is brought on by an aberrant accumulation of Cerebrospinal fluid in the brain's deep ventricles, or cavities (Azzam *et al.*, 2023). The ventricles enlarge as a result of the excess fluid, damaging pressure is applied to the brain's tissues. Executive function impairment can result from damage to the frontal lobes or from links with other areas and functions of the brain (Griffa *et al.,* 2020).

Electroconvulsive therapy: An electric current is passed to the brain during an ECT treatment (Thirthalli *et al.,* 2023). Retrograde amnesia is the side effect of this medication that lasts the longest (Othman *et al.,* 2021). Most patients experience memory lapses for events that happened shortly before or during the ECT shortly afterward, however the amnesia can last for months or even years.

Infections and Inflammatory Conditions: Infections that affect the brain, such as encephalitis, meningitis, or HIV/AIDS, can result in memory impairements, Inflammatory Conditions like multiple sclerosis can also affect memory function.

Brain Tumor and Head Injury: Memory can be impacted by any brain damage or disease (Turnquist *et al.,* 2020). Damage to the brain regions that make up the limbic system, which regulates emotions and memories, can lead to amnesia (Sparaco *et al.*, 2022).

Exposure to Heavy Metal: The hippocampal and frontal cortex, which are important for learning memory and executive processes, can become accumulated with heavy metals, including lead, cadmium, manganese, and vanadium (Bakulski *et al.*, 2020). Heavy metal buildup in these areas can directly affect cognitive functions. Exposure to vanadium has been connected to memory impairment and other memory issues. According to Ladagu et al. (2020), vanadium buildup in the brain causes apoptosis, DNA cleavage, and iron-mediated oxidative stress, which damages and malfunctions neurons.

2.5 VANADIUM NEUROTOXICITY

With an atomic number of 23, vanadium (V) is a transition metal that is widely found in nature. In 1801, Andres Manuel Del Rio became the first to report it by analyzing a new lead-bearing mineral he called "brown lead". However, Nils Sefstrom, a Swedish chemist, made the discovery in 1830 (Claudia *et al.*, 2022). The 22nd most common element in the earth's crust, this dazzling silver-white, soft, and pliable metal has drawn the attention of nutritionists due to its presence as a trace element in certain marine species (Enrique *et al.*, 2022). The main source of vanadium for the general public is ambient air (Rojas-Lemus *et al.,* 2021).

It is widely distributed in the air, although its function as a nutrition for humans is still unknown. Vanadium is mostly released into the environment by burning fuel oils that include vanadium, as well as fumes and dust (Celo *et al.,* 2021).

Vanadium's neurotoxic effects are still poorly understood. According to Quig (2020), animals that are acutely exposed through ingestion or inhalation may have changes in their neurological systems, paralysis of the legs, respiratory failure, convulsions, bloody diarrhea, and even death. Vanadium poisoning in mice results in disruption of the blood–brain barrier (kale *et al.,* 2020), changes in the concentrations of certain neurotransmitters, including dopamine, serotonin, and norepinephrine, as well as an inhibitory effect on norepinephrine uptake and release (Ahmed *et al.,* 2020).

Depending on their quantities, the biologically significant vanadium oxidation states vanadate (V5+) and vanadyl (V4+) are hazardous to mammals (Shaheen *et al.,* 2022). According to Majumder et al. (2020), employees who were exposed to vanadium pentoxide (V2O5) at work displayed changes in their cardiovascular system along with a range of symptoms related to their gastrointestinal, respiratory, and central neurological systems. Moreover, because elevated tissue vanadium levels lower serotonin concentration, it has been proposed that these levels may play a role in the etiology of manic-depressive illness. Additionally, people who were exposed to this metal on a long-term basis showed decreased cognitive capacities (Dobrakowski *et al.,* 2023).

2.5.1 VANADIUM SOURCES

The extraction of vanadium from coal or fossil fuels, such as Vanadium-rich coal tars and oil, explains the high V concentrations registered in the atmosphere (Brushett *et al.*, 2023). It is also found in phosphate rock, iron ores, and some crude oils in organic complexes and in small percentages of meteorites (Saleh*et al.,* 2021).

In metallurgy, vanadium is typically used in alloys with steel (joe *et al.,* 2021). Additionally, because it is nonferrous, it is essential to the atomic, space, and aviation manufacturing industries. In the chemical industry, Vanadium pentoxide and metavanadates are very significant for polymers and sulfuric acid synthesis. The enterprises that produce steel alloys may have elevated levels of heavy metal emissions nearby. According to Fortoul *et al.* (2021), vanadium is also released into the atmosphere during the resmelting of scrap steel, the conversion of titaniferous and vanadic magnetite iron ores into steel, the roasting of V slags, the V2O5 smelting furnaces, and the electric furnaces used to smelt ferrovanadium.

Ferrovanadium, vanadium pentoxide, vanadium trioxide, vanadium carbide, and salts like ammonium and sodium vanadate are the most important vanadium compounds. Powdered forms of the oxides and salts are utilized. According to reports, the metallurgical industry produces vapor that contains V2O5, which condenses to create an aerosol that is breathable.

2.5.2 VANADIUM ABSORPTION, DISTRIBUTION AND EXCRETION

Ferrovanadium is one of the most important vanadium compounds. According to Domingo et al. (2023), only 10% of ingested vanadium appears to be absorbed from the gastrointestinal tract. According to this study, the majority of ingested vanadium is converted in the stomach into the cationic vanadyl form before being absorbed in the duodenum by an unidentified mechanism (Yang *et al.*, 2019). Although the vanadate form also exists, vanadate is once more transformed into vanadyl ion once it enters the circulation (Banik *et al.*, 2020). Thus, blood proteins quickly deliver vanadate (by transferrin) and vanadyl (by albumin and transferrin) to different tissues (Levina *et al.*, 2023). The liver, kidney, brain, heart, muscles, and bone are among the organs and tissues that absorb vanadium after supplementation. It has been demonstrated that when mice are given parenterally administered Vanadium over an extended period of time, their kidney, spleen, bone, and liver tissues accumulate noticeably high levels of the metal (Enrique *et al.*, 2022). Unabsorbed Vanadium is excreted in feaces. Ten percent of the vanadium administered parenterally was discovered in the feces of rats and humans (Filler *et al.*, 2021). Urine and bile are the ways that vanadium is eliminated (Zwolak 2020).

Vanadium toxicity is contingent upon multiple parameters, such as the mode of administration and the toxicity of the V molecule. Vanadium is generally not very hazardous; its toxicity increases after parenteral injection and decreases after eating. One exposure method that results in intermediate toxicity is inhalation. Higher valences of V cause greater toxicity, with pentavalent compounds—such as vanadium pentoxide—generally having the highest levels of toxicity (Meng *et al.*, 2020).

2.5.3 VANADIUM EFFECTS IN THE NERVOUS SYSTEM

Vanadium is able to pass across the blood-brain barrier (Olopade *et al.,* 2019), and its compounds have the ability to affect neurologic function when administered in various ways (Jaiswal *et al.,* 2020). According to some reports, nursing rats exposed to vanadium experienced neurological abnormalities in their pups (Gilbert *et al.,* 2023); other authors documented neurological changes and an increase in brain Vanadium concentration following intraperitoneal treatment of sodium metavanadate (Folarin *et al.,* 2018). Neuroinflammation in the brains of mice exposed to V2O5 was also documented (Rojas-Lemus *et al.,* 2020).

Done conducted one of the earliest investigations into the neurological effects of V, discovering that people exposed to V showed signs of depression and tremor (Makova *et al.,* 2022). It has been shown by other investigations that those who are exposed to their jobs exhibit changes in their cognitive abilities (Ścibior *et al.,* 2023). Nerve cells and glia have been impacted by V exposure, regardless of the route, duration, and substance (Usende *et al.,* 2021). Folarin et al. found that the brain accumulates considerable levels of V, mostly in the brain stem, cerebellum, and olfactory bulb, after persistent intraperitoneal treatment at 3 mg/kg in mice. This study reported diminished apical dendrites in the hippocampus CA1, loss of pyramidal neurons, nuclear pyknosis, and alteration of the layering pattern in the prefrontal cortex. This study reported loss of cerebellar Purkinje cells, loss of pyramidal neurons, and diminished apical dendrites in the hippocampus CA1. It also described disturbance of the layering pattern in the prefrontal cortex with nuclear pyknosis. Microgliosis and astrogliosis accompanying these morphological changes.

Additionally, drinking milk from moms who were exposed to sodium metavanadate has been linked to demyelination (Manto *et al.,* 2021). Dorado-Martínez et al. also reported that Golgi staining demonstrated a significant loss in dendritic spines in the striatum compared to the controls in male CD-1 mice exposed to 0.02 M V2O5 2 hours twice a week for four weeks, indicating that the inhalation of V2O5 causes severe neuronal damage in this nucleus.

2.5.4 MECHANISMS OF VANADIUM NEUROTOXICITY

According to reports, vanadium causes the generation of reactive oxygen species (ROS), which some writers have suggested could be a plausible explanation for why it is neurotoxic (Dolan *et al.,* 2023). According to Wang et al. (2024), vanadium engages in the Fenton reaction together with other catalytic transition metals. In bodily fluids, V mostly occurs as V pentoxide (V_2O_5) in the 5+ oxidation state (Cohen 2022). V enters the cell by anion channels as vanadate, passive diffusion, and endocytosis attached to transferrin as vanadyl ions (Olaolorun *et al.,* 2021). Intracellular antioxidants convert vanadate to vanadyl upon entry into the cell, which leads to the generation of reactive oxygen species (ROS) (Geethika *et al.*, 2021). Then, in a Fenton-like reaction, hydrogen peroxide H_2O_2 oxidizes vanadyl into vanadate, producing hydroxyl radicals as a result (Cheng *et al.,* 2023). These processes cause oxidative stress and have harmful effects on proteins, lipids, and nucleic acids at higher V levels. V neurotoxicity is linked to myelin deficiencies because of the brain's high lipid content, which makes it susceptible to oxidant-induced lipid peroxidation (Femi-Akinlosotu *et al.,* 2022; Azeez *et al.*, 2020). Furthermore, as previously noted, previous findings indicated changes to the hippocampus (Ladagu *et al.,* 2023), disruption of the blood–brain barrier (Kale *et al*., 2020), and loss of substantia nigra tyrosine hydroxylase cells, which in turn caused dendritic spine loss in the striatum's medium-sized spiny neurons (Enrique *et al.,* 2022).

In addition to oxidative stress, V has been shown to be toxic to the cytoskeleton due to its ability to compete with phosphatases. As a result, V inhibits actin polymerization by inhibiting tyrosine phosphatases (Dolan *et al.,* 2023; Enrique *et al.,* 2024); this, in turn, disrupts microtubule formation and function by decreasing gamma-tubulin (Graff *et al.,* 2023). Actin polymerization is also well recognized to determine dendritic spine and dendritic dendrite morphology (Wayman *et al.*, 2023). These data lead us to speculate that inhaling V_2O_5 may trigger hippocampal cell death akin to that observed in Alzheimer's disease, the neurodegenerative condition that is the leading cause of dementia. Dementia symptoms range from issues with language, direction, and problem-solving to memory changes and other cognitive skill deficiencies that impair an individual's capacity to carry out daily tasks (Lazarus *et al.*, 2024). Disorientation and episodic and spatial memory loss are the most prominent signs at the start of the disease (Verde *et al.,* 2024). According to Song (2023), the hippocampus formation and associated cortices, which make up the medial temporal lobe region, are the first regions impacted in the evolution of the disease and are crucial for the proper operation of declarative and spatial memory systems.

2.6 GOTU KOLA

Gotu kola also known as Asiatic Pennywort is a common name for Centella asiatica which is a medicinal herb (Yousaf *et al.,* 2020). The species is valued for its medicinal and nutritional properties. Watanabe et al. (2021) describe it as a perennial plant with edible leaves and stems that resemble green leafy vegetables.

The taxanomical classification of the plant is as follows (Xian *et al.,* 2020);

Kingdom: Plantae

Phylum: Magnoliophyta

Class: Magnoliopsida

Subclass: Rosidae

Order: Apiales

Family: Apiaceae

Genus: Centella

Specie: Centella asiatica

Its therapeutic effects are numerous and it has been widely used in the East as a panacea for all ailments (Watanabe *et al.,* 2021). The species is described as a low-growing, creeping, perennial herbaceous plant that flowers, roots at nodes, and has prostrate or semi-erect stems. Its height ranges from 10.0 to 45.0 cm (Parker, 2020). At the nodes, the shovel-shaped, reniform leaves emerge in alternating rosettes (Ravi *et al.,* 2019). Being an entomophilous plant, it can procreate vegetatively by sending out long runners or by using seeds.

The nutritional value, medicinal use, and biological activity of the herbaceous plant have garnered increased interest in recent times. Because of its extensive range of advantageous neuroprotective activities, Centella asiatica has been regarded as a brain tonic. In addition, a number of other effects have been documented, including wound healing, anti-inflammatory, antiproliferative, anticancer, antioxidant, and antiulcer properties. Numerous phytochemicals, including asiactic acid, asiatosides, polyphenolic compounds, etc., have been shown to be beneficial for these responses in animal models used to evaluate these effects (Andola *et al.,* 2019).

2.6.1 PHYTOCHEMICAL COMPOSITION

Asiaticoside, brahmoside, asiatic acid, and brahmic acid (madecassic acid) are among the trisaccharide derivatives and pentacyclic triterpenoids found in Centella Asiatica (Johnson *et al.*, 2022). Centellose, centelloside, and madecassoside are additional components (Mitra *et al.*, 2021). From Centella asiatica, about 124 chemical compounds have been separated and identified (Torbati *et al.*, 2021). There are 5.4% insoluble dietary fiber, 0.49% soluble dietary fiber, and 87.7% moisture in fresh leaves. According to Shukurova *et al.*, the plant has low levels of fats (0.2%), proteins (2.4%), and carbohydrates (6.7%), but high levels of vitamins, including vitamin B1 (0.09 mg/100 g), vitamin A (738 IU), vitamin C (7 mg/100 g), and minerals like calcium (171 mg/100 g), P (32 mg/100 g), K (468.59 mg/100 g), and Fe (5.6 mg/100 g). Numerous conducted studies have pointed out that major bioactive constituents attributing to its medicinal value are triterpenes (asiaticoside, asiatic acid, madecassic acid, madecassoside), (Ghorai *et al.*, 2023), which have also been regarded as its biomarker components (Jusril *et al.*, 2020).

2.6.2 GOTU KOLA AS A NEUROPROTECTIVE AGENT

About 9·0 million fatalities globally in 2016 were caused by neurological illnesses (including deaths of people with disabilities). Effective advances in medicine and therapeutic approaches are urgently needed, as the global burden of these diseases rises with aging populations and presents a growing challenge for rehabilitation and treatment (Feigin *et al.*, 2020). The neurotherapeutic advantages of madecassoside and asiaticoside have been the subject of the most research. These two active ingredients that were isolated from C. asiatica have been shown to improve memory and learning, protect the brain against neurodegenerative diseases and cognitive deficiencies, lessen anxiety and depressive symptoms, and provide general nervous system protection. (Mandal *et al.*, 2023).

For neurotherapeutic medications to be clinically beneficial, they need to be able to pass through the blood-brain barrier (BBB). The blood-brain barrier (BBB) regulates blood flow to and from the brain in order to preserve brain homeostasis and prevent the entry of harmful substances. It is a continuous endothelial membrane found within the brain micro-vessels where brain endothelial cells are sealed by tight junctions (Abdullah *et al.*, 2021).

The etiology of several neurological disorders, including Alzheimer's disease (Song *et al.*, 2020) and traumatic brain and spinal cord injuries (Sulhan *et al.*, 2020), is linked to disruptions of the blood-brain barrier. Therapeutics that can cross and interact with the BBB sufficiently are necessary for the treatment of CNS disorders. Rather from being useless due to insufficient potency, many medications are rendered ineffective against neurological illnesses because they are unable to cross the blood-brain barrier (BBB) (Liang *et al.*, 2019). In a study, Hanapi, Nur Aziah, et al. used primary pig brain endothelial cells (PBECs) as a model to examine the degree of BBB permeability by C. asiatica chemicals. The results of the experiment showed that the examined phytocompounds had an exceptionally high ability to pass the blood-brain barrier (BBB), with madecassoside and asiaticoside exhibiting the highest permeability. Notably, the compounds also showed higher permeability coefficient values than donepezil, a drug commonly used for AD. (Hanapi *et al.*, 2021). Neuroprotective benefits of Gotu kola include;

Neurogenesis: Gotu Kola promotes the development of new brain cells. It promotes dendritic branching, which increases neuronal transmission and brain plasticity (Hannan *et al.*, 2021).

Neuroprotection: The brain is shielded from toxins and oxidative stress by gotu kola. Research demonstrates that it aids in defense against food additives and heavy metals, which can trigger migraines, mood swings, and cognitive fog. Additionally, it prevents lipid peroxidation, which can harm cells and be a factor in a number of disorders, and lowers oxidative stress by lowering free radicals in brain cells. Additionally, they suppress the synthesis of reactive oxygen species and increase the activity of antioxidant enzymes like superoxide dismutase, which support cellular homeostasis and lessen oxidative damage (Eduviere *et al.*, 2022). Additionally, it dramatically lowers glutathione and glutathione peroxidase levels as well as the synthesis of glutamate-induced nitric oxide (NO). Additionally, there is also an Inhibitory activity against phospholipase A2 (PLA2), which is involved in neuropsychiatric disorders.

Acetylcholinesterase inhibition: The cholinergic enzyme acetylcholinesterase (AChE) is mostly located in postsynaptic neuromuscular junctions, particularly in nerves and muscles. Acetylcholine is broken down by the body's principal cholinesterase enzyme into acetic acid and choline. Its efficacy is limited because this breakdown breaks down the nerve cells' ability to communicate with their target. This enzyme is inhibited by gotu kola, which blocks it and causes acetylcholine to accumulate in the synapses and cholinergic receptors to continuously activate (Gofir *et al.*, 2021).

Glutamic acid decarboxylase booster: The enzyme glutamate decarboxylase, also known as glutamic acid decarboxylase (GAD), catalyzes the conversion of glutamate into carbon dioxide (CO2) and gamma-aminobutyric acid (GABA). It has been discovered that gotu kola dramatically (more than 40%) increases the activity of the Glutamate Decarboxylase (GAD) enzyme, which raises GABA levels in the brain.

Tyrosinase regulation: High dopamine production and, by extension, high neuromelanine production can result from elevated tyrosinase activity. Protein proteostasis is disrupted by neuromelanine buildup. Tyrosinase activity is regulated by gotu kola, which in turn regulates dopamine levels.

Enhancement of cyclic Adenosine Monophosphate (cAMP): By increasing the phosphorylation of cyclic AMP response element binding protein (CREB) in neuroblastoma cells in Aβ(1-42) proteins found within the amyloid plaques that form in AD patients' brains, gotu kola also exhibits neuroprotection. [57].

2.6.3 SIDE EFFECTS OF GOTU KOLA

Although they are uncommon, side effects can include headaches, upset stomachs, nausea, dizziness, burning sensations on the skin, and excessive sleepiness. These usually occur when gotu kola dosages are high.

2.7 OTHER TREATMENT USE IN MEMORY IMPAIRMENT

- Cholinesterase inhibitors

These include drugs such as Donepezil, Rivastigmine, Galantamine. These drugs works by blocking the cholinesterase enzyme that breaks down acetylcholine which is a neurotransmitter that plays a vital role in memory and learning, thereby causing a build-up of acetylcholine in the synapse. Though generally well-tolerated, if side effects occur, they commonly include nausea, vomiting, loss of appetite and increased frequency of bowel movements.

- Glutamate regulators

Drug such as Memantine are classified as glutamate regulator and they help control the amount of glutamate in the central nervous system to an optimal level. Memantine is an antagonist of the N-methyl-D-aspartic acid (NMDA) receptor that prevents calcium from entering neurons and inflicting damage to nerves. Side effects includes headache, constipation, confusion and dizziness.

- Cholinesterase inhibitors + Glutamate regulators

This kind of medication combines a glutamate regulator with a cholinesterase inhibitor. An example is donepezil and memantine which is available under the brand-name prescription drug Namzaric.

- Amyloid inhibitors

These include Aducanumab and Lecanemab. They acts by destroying plaques of toxic protein known as beta-amyloid. This is the protein that researchers believe plays a role in the cognitive decline of people with Alzheimer's disease. Serious allergic responses may result from these therapies. Amyloid-related imaging abnormalities (ARIA), infusion-related responses, headaches, and falls are other possible side effects.

- Ginko biloba supplement (GB-S)

It is a neuroactive supplement which has been reported to demonstrate neuroprotective effects. GB-S significantly reduced lipid peroxidation and nitrite level, inhibited TNF-α and acetylcholinesterase activity and improved glutathione, catalase and superoxide dismutase activity in the hippocampus. It also inhibits hippocampal apoptosis via elevated expression of caspase-3 with marked increase level of brain derived neurotrophic factor (BDNF). (Ben-Azu *et al.*, 2022).

CHAPTER THREE

3.0 MATERIALS AND METHODS

3.1 EXPERIMENTAL ANIMALS

Male Albino Swiss mice weighing 24.0±2.0 g was used for this research study. Subjects were collected from the animal house, college of basic medical sciences, Delta State University, Abraka and were housed in plastic cages at room temperature with 12:12 h light–dark cycle. They were fed with high quality commercial rodent feed and water thrice a day. Mice were acclimatized for at least one week before the start of the experimental research. The experimental procedures were performed in accordance with the National Institute of Health (NIH) Guideline for the Care and Use of Laboratory Animals (Ahmadi-Noorbakhsh *et al.*, 2021).

3.2 EQUIPMENT AND APPARATUS

The equipment used in this study includes; Centrifuge (ATKE), water bath (Equitron), spectrophotometer (Inesa, 752N), pH meter (EDT instruments), weighing balance (Ohaus), test tubes, eppendorf tubes, tube racks, dissection kits, boards, towels, tissue paper, petri dish, oral canula, permanent marker, stop watch, beaker, timer, ice and refrigerator, cotton wool, aluminium foil, needles and syringes, universal sample bottles, hand gloves, nose masks, paper tapes , measuring cylinders, homogenizer, spatula, feeding trough and drinker.

3.3 DRUGS AND CHEMICALS

Gotu kola (Centella Asiatica), Vanadium pentoxide (V_2O_5) 5,5′-dithio-bis(2-nitrobenzoicacid)-DTNB (Aldrich, Germany), trichloroacetic acid-TCA (Burgoyne Burbidges & Co., Mumbai, India), thiobabituric acid-TBA (Guanghua Chemical Factory Co. Ltd., China), Tris (hydroxymethyl)-amino-methane (Tris-buffer) (Hopkin & Williams Company, USA), Acetic acid (Sigma-Aldrich, Inc., St Louis, USA), $NaHCO_3$ (BDH Chemicals Ltd, Poole, England), Sodium Carbonate (fisons, Loughborough Leics, England), $Na_2HPO_4.H_2O$ (BDH Chemical Ltd, Poole, England), $NaH_2PO_4.H_2O$ (BDH Chemical Ltd, Poole, England), K_2HPO_4 (BDH Chemical Ltd, Poole, England), $K_2Cr_2O_7$ (BDH Chemical Ltd, Poole, England), KCl (BDH Chemical Ltd, Poole, England), NaOH (J.T Baker Chemicals Co., Phillipsburg, N.J., USA) were used in the study.

3.4 DRUG PREPARATION AND TREATMENT GROUPS

Briefly, 0.03g of Vanadium was dissolved in 30ml of distilled water to obtain a stock solution of 10mg/ml concentration. 435mg of Gotu kola as contained in a capsule was dissolved in 43.5mL of distilled water to obtain the stock solution 10mg/ml. Male mice received oral (p.o) doses of Gotu kola (50mg/kg and 100mg/kg), or distilled water (10 mL/kg), via the use of oral cannula. The dose of Vanadium used in the study were selected based on the results obtained from pilot studies performed in our laboratory. Thus, the animals were allotted into four (4) treatment groups (n = 6): groups 1 received distilled water (10 mL/kg p.o) while groups 2 received Vanadium 10mg/kg, group 3 and 4 received Gotu kola (50mg/kg and 100mg/kg p.o respectively) and Vanadium (10mg/kg Intraperitoneally). The time of drug administration was between the hours of 9:00 to 10:00 am each day.

3.5 EXPERIMENTAL DESIGN

The mice were randomly distributed into 4 groups (n = 6); group I (positive control) were given vehicle (distilled water 10 mL/kg), group 2 (negative control) were given Vanadium (10 mg/kg) while groups 3 and 4 received Gotu kola (50mg/kg and 100mg/kg respectively), orally. One hour later, mice in groups 3 and 4 were further given Vanadium base on their body weight (10mg/kg). The animals were treated for 14 days and on the 14th day, behavioural tests (Y-maze use in assessment of spatial working memory and the open field chamber use in the assessment of recognition memory) were carried out between the hours of 9:00 am and 12:00 noon.

3.6 BEHAVIOURAL TEST

3.6.1 Assessment of spatial working memory using the Y-maze

The Y-maze was used to assess short term memory in mice. Spontaneous alternation, a measure of spatial working memory, was assessed by allowing mice to explore all three arms of the maze. This test is driven by an innate curiosity of rodents to explore previously unvisited areas (Sarnyai *et al.*, 2019). Each mouse was placed in the center of the maze and allowed to explore the arms for 5 min using a timer. All arm entries were recorded accordingly except the arm that the mouse entered when it was first placed in the maze. The arm entry was scored when the four paws of the animals were completely in the arm of the Y-maze.

Thereafter, the percentage alternation which is a measure of spatial working memory was calculated using the formula:

(Total alternation number/Total number of entries − 2) $\times$ 100.

The maze was cleaned with 70% alcohol to eliminate possible bias due to odor cues.

3.6.2 Assessment of recognition memory using the open field chamber

The effect of Vanadium on memory performance was assessed using the Novel Object Recognition (NOR) test in an open-field chamber (60 cm × 50 cm × 40 cm) as previously described (Wang *et al.*, 2009) with objects (A, B and C) identically sized (4.5 cm diameter and 11.5 cm height) cylindrical bottles. Objects A and B were white, whereas the discriminate object C had a black and white pattern. The NOR test consists of two phases; the trial phase and the test phase. The trial phase was carried out by placing each mouse in the middle of two identical objects (A and B) on opposite sides (at a distance of 8 cm from the walls and 34 cm from each other) of the open-field chamber for 5 min and the duration of time spent (s) in exploring each of the objects was recorded. Thereafter, the animals were returned to their home cages for an interval of 2 hrs. In the test phase, object B was replaced with object C, which was novel to the mice and different from either object A or B. Mice were then left to explore objects A and C for a period of 5 min. The time spent exploring both familiar and novel objects were recorded. Exploration recorded were times the mouse's nose is pointed toward the object and within 2–3 cm of the object, with active sniffing. The objects and the arena were cleaned after every individual mouse session with 70% alcohol to minimize odor cues. The discrimination index/percentage preference, which was used as a measure of recognition memory function, was calculated as the difference in time exploring the novel and familiar objects divided by the total amount of time spent with both objects (Wang *et al.*, 2009).

3.7 PREPARATION OF BRAIN TISSUES FOR BIOCHEMICAL ASSAYS

After the behavioural tests, mice in the respective groups were euthanized and the brains were rapidly removed. Thereafter, each mouse brain obtained was weighed, homogenized with 10% w/v phosphate buffer (0.1M, pH 7.4) and centrifuged for 10,000 rpm at 4oC for 15 min. Then, the supernatants of each brain were used for the various biochemical assays.

3.7.1 Determination of acetylcholinesterase (AChE) activity in mice brain

Aliquots of supernatant of individual mouse brain of the various treatment groups were taken and used to measure AChE activity, a marker for cholinergic neurotransmission (Ellman *et al.,* 1961). Briefly, AChE activity in the homogenate was measured by adding 2.6mL of phosphate buffer (0.1 M, pH 7.4), 0.1 mL of 5,5′-dithio-bis(2-nitrobenzoic acid) (DTNB) and 0.4 mL of the homogenate. Then, 0.1 mL of acetylthiocholine iodide was added to the reaction mixture. The absorbance was read using a spectrophotometer at a wavelength of 412 nm and change in absorbance for 10 min at 2 min interval was recorded. The rate of AChE activity was measured by following the increase in colour produced from thiocholine when it reacts with DTNB. The change in absorbance per minute was determined and the rate of AChE activity was calculated and expressed as µmol/min/g tissue.

3.7. 2 Determination of glutathione peroxidase (GPx) activity

The activity of GPx in the brain tissue homogenates was measured using the spectrophotometer method described by Flohé and Günzler (1984). Homogenizing brain tissue samples involved adding 1 mM EDTA to 0.1 M (pH 7.0) phosphate buffer. The homogenate was then centrifuged at 10,000g for 10 minutes at 4ºC in order to extract the supernatant. The reaction mixture contained 0.1 mL of the supernatant sample, 0.7 mL of phosphate buffer, 1 mM reduced glutathione solution, 1 mM NADPH solution, and 0.1 mL glutathione reductase (GR) solution. The procedure began with the addition of 0.1 mL of 0.25 mM hydrogen peroxide (H2O2). At 25ºC, the absorbance wavelength was measured with a spectrophotometer at 340 nm. Absorbance values were measured every minute for five minutes. The GPx activity was calculated based on the rate of NADPH oxidation, and it was discovered that the extinction coefficient of NADPH at 340 nm was 6.22 $mM^{-1} cm^{-1}$. The amount of enzyme needed to catalyze the oxidation of one micromole of NADPH per minute is known as one unit of GPx activity. The unit of measure for the activity was unit per milliliter (U/mL).

3.8 PREPARATION OF BRAIN TISSUES FOR HISTOLOGY

Brain tissue obtained for histological studies from the mice in each group was fixed in 10% phosphate buffered formaldehyde used for perfusion. The transverse sections (5–6 µm thick) of the Cerebral cortex with subcortical white matter were then obtained using a microtome (Leica, Germany) and processed by the routine method for paraffin wax embedment for histology. The histological staining with H&E was carried out in paraffin wax embedded sections for cell quantification as earlier described by Emokpae *et al.* (2020) and the sections were fixed on glass slides for microscopy and photomicrography.

3.9 HISTOLOGY AND ESTIMATION OF NEURONAL DENSITY

Viable neuronal cells, defined as round-shaped, intact cytoplasmic cells, without any distortions (Emokpae *et al.,* 2020) were counted within a given square area of the circular view in a section with the aid of a binocular microscope.

3.10 STATISTICAL ANALYSIS

The data obtained were presented as Mean ± SEM. The result were analysed by one way analysis of variance (ANOVA) and followed by Student's Newman–Keuls post hoc test. Graph Pad InStat® Biostatistics software was used to determine the level of significant for all tests which was set at $p < 0.05$.

CHAPTER 4

4.0 RESULTS

4.1 Effect of Gotu kola on Vanadium induced memory impairment in mice utilizing the Y-maze test

The effect of Gotu Kola on spatial working memory of mice exposed to Vanadium for 14 days using the Y-maze test is shown in fig 4.1. Using one way ANOVA and post hoc comparison, the figure showed that there was a significant ($p < 0.05$) reduction in the percentage alternation of mice exposed to vanadium as compared to the control group which is an indication of memory impairement. However, on administration of Gotu Kola (50mg/kg p.o) there was significant ($p < 0.05$) increase in the percentage alternation as compared to mice treated with vanadium, and Gotu kola (100mg/kg p.o) also showed similar increase in percentage alternation.

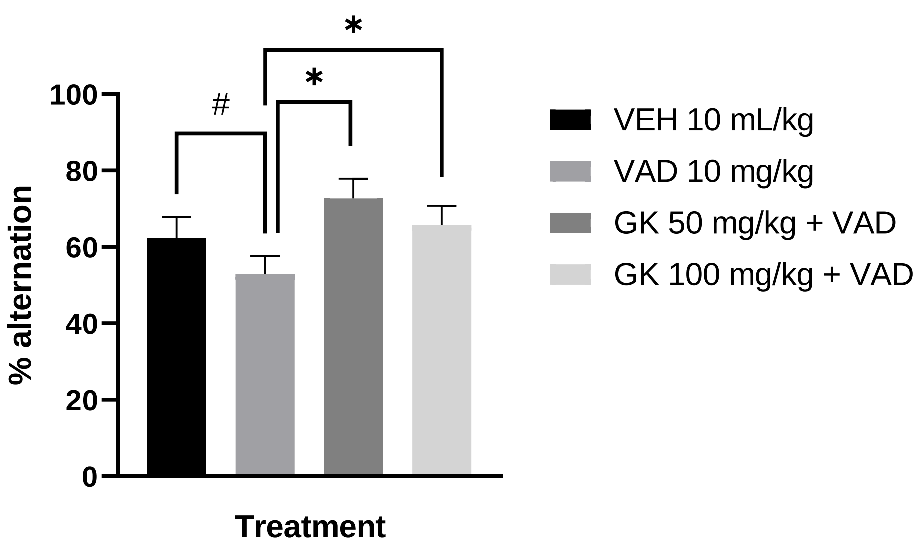

Figure 4.1: Effect of Gotu kola on Vanadium induced memory impairment in mice utilizing the Y-maze test

Values represent the mean ± S.E.M for 6 animals per group

# p<0.05 compared to the control group (ANOVA followed by Turkey's post hoc test).

* p<0.05 compared to the pathologic group (ANOVA followed by Turkey's post hoc test).

VEH ---- Vehicle, **GK** ---- Gotu Kola, **VAD** ---- Vanadium

4.2 Effect of Gotu kola on Vanadium induced memory impairment in mice utilizing the Novel object recognition test

The effect of Gotu kola on recognition memory of mice exposed to Vanadium for 14 days using the Novel Object Recognition Test (NORT) is shown in Fig 4.2. As shown in the figure, there was significant ($p<0.05$) decrease in discrimination index in mice treated with vanadium as compared to the control group which indicates a decrease in preference for novel Object and a biomarker of cognitive deficit. However, Gotu Kola (50mg/kg and 100mg/kg p.o) significantly ($p<0.05$) increased the discrimination index compared to mice exposed to Vanadium only.

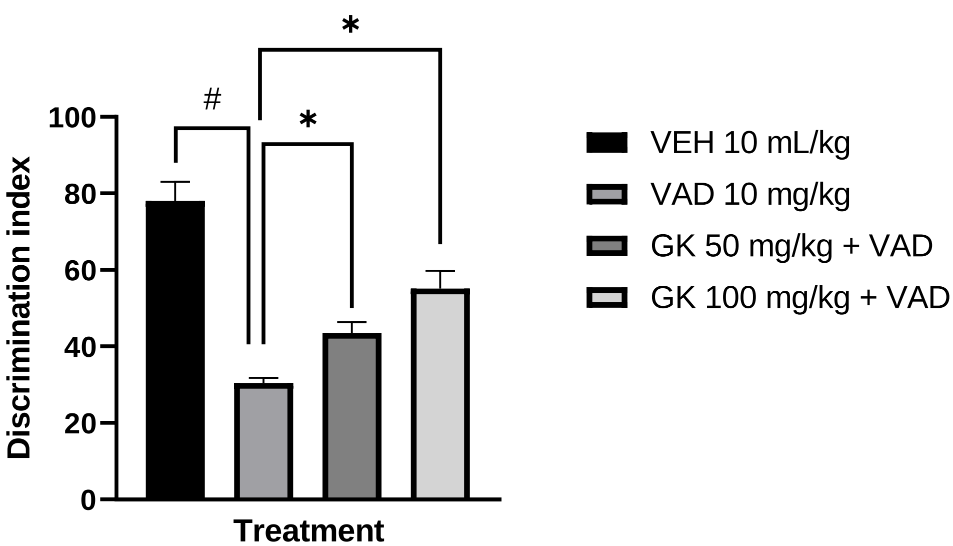

Figure 4.2: Effect of Gotu kola on Vanadium induced memory impairment in mice utilizing the Novel object recognition test

Values represent the mean ± S.E.M for 6 animals per group

# $p<0.05$ compared to the control group (ANOVA followed by Turkey’s post hoc test).

* $p<0.05$ compared to the pathologic group (ANOVA followed by Turkey’s post hoc test).

VEH ---- Vehicle, **GK** ---- Gotu Kola, **VAD** ---- Vanadium

4.3 Effect of Gotu kola on Vanadium induced memory impairment in Acetylcholinesterase activity in brain

The effects of Gotu kola on brain Acetylcholinesterase (AChE) activity of mice exposed to Vanadium is shown in fig 4.3. As shown, there is significant ($p<0.05$) increase in brain AChE activity in mice treated with vanadium as compared to the control/vehicle group. However, Gotu kola (50mg/kg and 100mg/kg p.o) significantly ($p<0.05$) decrease the brain AChE as compared to the pathologic group i.e mice treated with vanadium only.

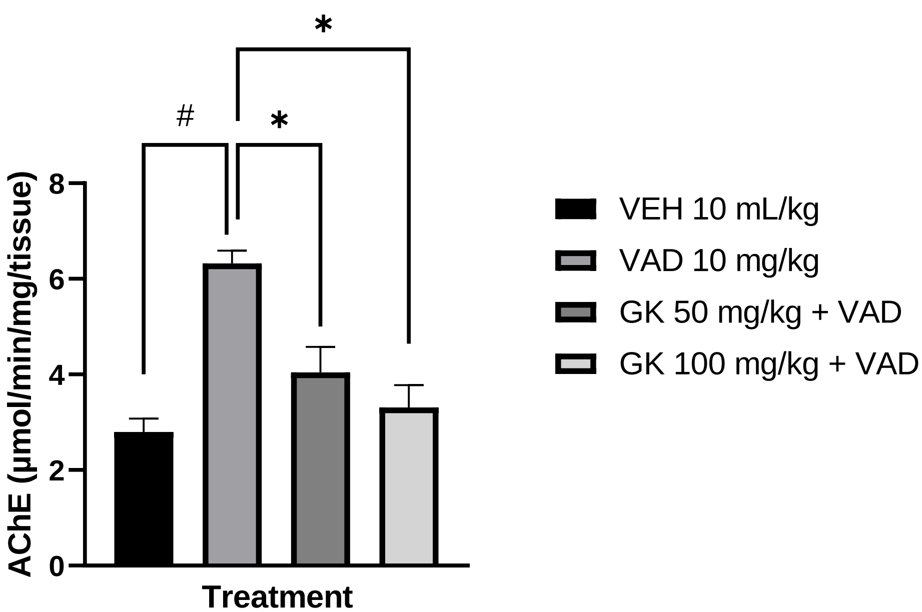

Figure 4.3: Effect of Gotu kola on Vanadium induced memory impairment in Acetylcholine in brain

Values represent the mean ± S.E.M for 6 animals per group

# $p<0.05$ compared to the control group (ANOVA followed by Turkey's post hoc test).

* $p<0.05$ compared to the pathologic group (ANOVA followed by Turkey's post hoc test).

VEH ---- Vehicle, **GK** ---- Gotu Kola, **VAD** ---- Vanadium

4.4 Effect of Gotu kola on Vanadium induced memory impairment in Glutathione peroxidase in brain

The effect of Gotu Kola on brain glutathione peroxidase activity of mice exposed to Vanadium is shown in fig 4.4. As shown, there is significant ($p<0.05$) decrease in brain glutathione peroxidase activity in mice treated with vanadium as compared to the control/vehicle group. However, Gotu Kola (50mg/kg and 100mg/kg p.o) significantly ($p<0.05$) increased the brain glutathione level as compared to Vanadium treated mice as the dose of Gotu kola was increased.

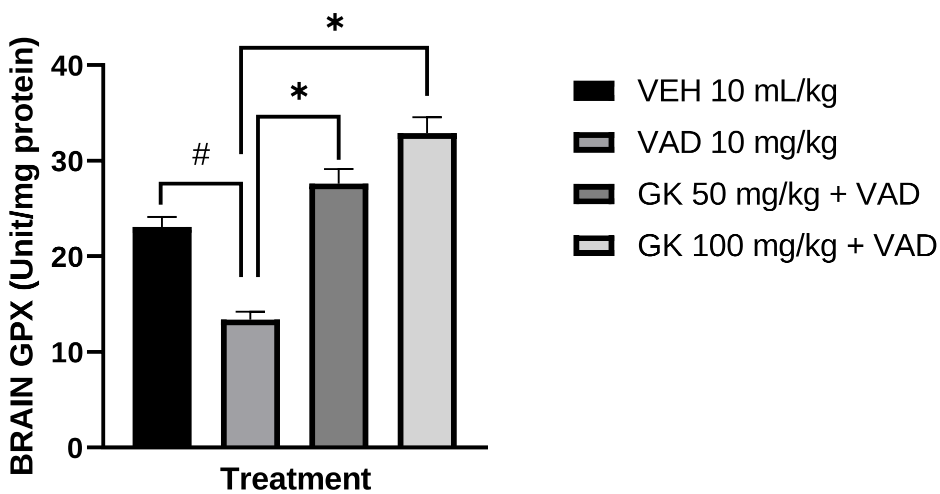

Figure 4.4: Effect of Gotu kola on Vanadium induced memory impairment in Glutathione peroxidase in brain

Values represent the mean ± S.E.M for 6 animals per group

# $p<0.05$ compared to the control group (ANOVA followed by Turkey's post hoc test).

* $p<0.05$ compared to the pathologic group (ANOVA followed by Turkey's post hoc test).

VEH ---- Vehicle, **GK** ---- Gotu Kola, **VAD** ---- Vanadium

CHAPTER 5

5.0 DISCUSSION, CONCLUSION AND RECOMMENDATION

5. 1 Discussion

From the results obtained during the experimental procedure, it showed Vanadium caused memory deficits as evidenced by a decreased in alternation behavior of mice in Y-maze paradigm. This result is in agreement with previous studies, which showed that heavy metals like Vanadium produced learning and memory deficits in rodents(Usende *et al.,* 2022). The Y-maze test is a recognized animal model routinely used for the evaluation of memory performance in rodents. The validity of the Y-maze model in assessing spatial working memory is based on the ability of rodents to visit any of the three arms (A, B, C) of the Y-maze, while trying to remember the correct sequence of arm alternation (Dalkiran *et al.,* 2021). It is assumed that the animal remember entering a previous arm and has preference for a novel arm, thus presenting as a measure of memory referred to as spontaneous alternation (Habedank *et al.,* 2021). The list of arm explored is believed to be held in spatial working memory, serving as a measure of short-term memory.

Vanadium also affected recognition of mice in the pathologic group (group 2) as there was a significant decline during Novel Object Recognition Test (NORT) while animals treated with Gotu Kola increased object recognition. The Novel Object Recognition Test (NORT) is used to evaluate cognition, particularly recognition memory, in rodent models of CNS disorders. This test is based on the spontaneous tendency of rodents to spend more time exploring a novel object than a familiar one. The choice to explore the novel object reflects the use of learning and recognition memory.

Data from the result showed a significant increase in brain Acetylcholinesterase (AChE) activity after mice were treated with vanadium for 14 days. Acetylcholinesterase is an enzyme involve in the termination of cholinergic activity by breaking down acetylcholine neurotransmitter. The significant increase in AChE activity in the brain of the present study therefore contributes to the effect of Vanadium toxicity in the brain. Administration of Gotu Kola significantly reduced the activity of AChE in Vanadium treated mice.

Vanadium also reduces brain glutathione level as seen in the result in Fig 4.4 thereby inducing oxidative stress and inflammatory biomarkers (Rojas-Lemus *et al.,* 2020). Glutathione is a powerful antioxidant that plays a crucial role in protecting cells, including those in the brain, from oxidative stress and damage. Mice treated with Gotu kola showed an increase in glutathione peroxidase, therefore it evidently shows the antioxidant activity of Gotu Kola.

5.2 Conclusion

In conclusion, the result of this study provides an evidence that Gotu Kola mitigates memory impairement and neuronal damage caused by heavy metals such as Vanadium through the inhibition of acetylcholinesterase activity, increase in the level of antioxidants and increase glutathione peroxidase.

This study therefore suggest that Gotu Kola has therapeutic effect in the management and treatment of memory disorders.

5.3 Recommendation

This study recommends that further investigations should be done to elucidate the beneficial role of Gotu Kola in memory disorders in human subjects and to validate the safety and efficacy of Gotu kola.

Further study should also be done to investigate the synergistic effect of Gotu kola with other antioxidants or detoxifying agents which could enhance its therapeutic efficacy against heavy metals toxicity.

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